

EDUCATION

- University of Edinburgh** Edinburgh, UK
 - PhD in Computer Science* *August 2023 - Present*
 - Research Field: AI-Systems, Distributed ML, ML-oriented Architecture, Serverless*
- University of Science and Technology of China** Hefei, China
 - Bachelor of Computer Science; The school of Gifted Young;* *July 2017 - June 2021*
 - Courses: Operating Systems, Artificial Intelligence, Principles of Compiler, Computer Architecture, High Performance Architecture*

PUBLICATIONS

- Yeqi Huang**, Congjie He, Haocheng Xiao, et al. "MICA: An End-to-End Compiler Stack for Mesh Accelerators." *Under Review, OSDI 2026.*
- Yeqi Huang**, Yanwei Ye, Guomin Chen, et al. "SwarmX: Agentic Cluster Scheduling Framework." *Under Review, OSDI 2026.*
- Yeqi Huang**, Yue Chen, Yanwei Ye, et al. "BioVLM: Evidence-Enriched Data Synthesis from Biomedical Papers." *Under Review, ACL 2025 System Demonstration.*
- Chuanming Zha, Jiamin Zheng, **Yeqi Huang**, et al. "LLM-Monitor: Efficient Privacy Violation Monitoring for LLMs." *Under Review, ACL 2025 System Demonstration.*
- Yinsicheng Jiang*, **Yeqi Huang***, Liang Cheng, et al. "ContextPilot: Efficient Retrieval-Augmented Generation with Accuracy-Preserving Context Reuse." *MLSys*, 2026. (*Equal Contribution)
- Congjie He, **Yeqi Huang**, Pei Mu, et al. "WaferLLM: Large Language Model Inference at Wafer Scale." *Under Review, OSDI 2025.*
- Yao Fu*, Yinsicheng Jiang*, **Yeqi Huang***, et al. "MoE-CAP: Cost-Accuracy-Performance Benchmarking for Mixture-of-Experts Systems." *NeurIPS*, 2025. (*Equal Contribution)
- Yao Fu, Leyang Xue, **Yeqi Huang**, et al. "ServerlessLLM: Locality-Enhanced Serverless Inference for Large Language Models." In *18th USENIX Symposium on Operating Systems Design and Implementation (OSDI 24)*, 2024.
- Xiao-Long Chen, Lin-Feng Wang, **Yeqi Huang**, et al. "Symplectic structure-preserving particle-in-cell whole-volume simulation of tokamak plasmas." In *SC21: International Conference for High Performance Computing, Networking, Storage and Analysis*, 2021.

PROJECTS

- QED Simulation on GPU:**
 - Implemented high-performance Monte-Carlo simulation achieving 1500x to 5000x speedup.
 - Received recognition from Nobel Laureate Frank Wilczek at APS conference.
 - Demonstrated expertise in CUDA programming and GPU micro-architecture optimization.
- AutoReader Project:**
 - Built a vector database for the latest ArXiv papers for daily semantic search and subscription.(Full-stack AI application)
 - Achieving **60x faster indexing than NVIDIA cuVS** library and **13x faster retrieval than Milvus**.
 - Extended system to support bioRxiv and PubMed papers with specialized biology-focused features.
 - Biology patched version is demonstrating in Dartmouth university, will submit to Nature this year.
- MICA (submitted to OSDI 2026):**
 - Developed the first end-to-end compiler stack for mesh accelerators (Cerebras, Tenstorrent) with a novel MeshIR.
 - Implemented a search framework reducing schedule search from weeks to hours for millions of cores.
 - Achieved 3.6x performance improvements over state-of-the-art compilers and 1.4x over expert hand-tuned codes.
 - Enabled applications to run 100-200x faster than on GPUs by efficiently exploring temporal-spatial schedules.
- SwarmX (submitted to OSDI 2026):**
 - Presented the first scheduler agent framework that formulates cluster scheduling as an agentic intelligence problem.

- Deployed in production on tens of thousands of servers, improving P99 latency by up to 50% and doubling throughput.
- Designed modular agents with neural predictors and memory to optimize complex workflow performance.
- Maintained compatibility with Ray and ComfyUI, enabling existing workflows to run without modification.
- **BioVLM (submitted to ACL 2025 System Demonstration):**
 - Automated pipeline transforming biomedical papers into evidence-enriched VLM training data.
 - Progressive post-training (SFT + RL) on Qwen3-VL-8B, training BioVLM-8B at under \$200 cost.
 - Achieved 48.0% weighted accuracy on LAB-Bench, 12.6% above base model and 3.8% above GPT-5.2.
- **LLM-Monitor (submitted to ACL 2025 System Demonstration):**
 - Designed a low-cost, high-quality, evolvable privacy violation monitoring system for LLMs.
 - Organized and compiled monitoring requirements into optimized representations executable by local medium-scale models.
 - Achieved 15× faster monitoring than GPT-5.2 using a specialized Qwen3-8B-based model.
 - Matched large public model (GPT-5.2) monitoring quality while being significantly faster and cheaper.
- **AI4Math - Sketchpad Project:**
 - Received grant from AI for Math Fund to develop Sketchpad system for formal mathematics (announcement)
 - Building system that automatically converts mathematical proofs into structured data using graph-based representation.
 - Enabling decomposition of proofs into individual statements for more precise auto-formalization.
- **More Projects (<https://yeqi-huang.com/>):**
 - I have plenty of AI related projects on my personal page.

TALKS

- 1 **Yeqi Huang.** "InfiniTensor: A Tensor-Friendly, Efficient Parallel Programming Library for Accelerator-Centric Clusters." *UKSys 2024*.
- 2 **Yeqi Huang.** "Why we need a new clustering benchmark in AI retrieval?" *International Workshop on Efficient Generative AI*, 2024.

TEACHING EXPERIENCE

- **Machine Learning System TA (2025):**
 - Teaching assistant for CUDA programming course, covering GPU architecture, CUDA, CuPy, and Triton.
 - Designed hands-on exercises and projects to help students master parallel programming concepts.
- **Machine Learning System TA (2026):**
 - Teaching assistant with new CuTile programming module, teaching Tile-based GPU programming.

AWARDS

International Awards:

- International Supercomputing Conference Student Cluster Competition Champion - 2021
- Asian Supercomputer Conference Student Cluster Competition First Prize - 2021

National Awards:

- Best Chinese Supercomputing Application of the Year - 2022
- Huawei Pioneer Developer (4 in China) - 2021
- National Compiler Designing Competition Second Prize - 2021

SKILLS

- **AI-Related:**
 - In-depth understanding of LLMs, with experience porting high-performance inference and training frameworks to various hardware platforms, including Apple Silicon, GPUs, and Cerebras.
 - Strong understanding of vector retrieval, having developed and tested graph and vector databases on multiple platforms.
 - In-depth knowledge of Multi-Agent applications and developed efficient development components for such applications.
- **Computer Science Related:**
 - Highly skilled in **C++**, **Python**, **Rust**; familiar with Go, JavaScript, and Latex
 - Proficient with **CUDA**, **Intel ONEAPI**, **OpenMP** and related parallel and distributed programming
 - Well-versed in **LLVM**, frequently participating in LLVM Forum online discussions
 - Extensive experience working with Linux, including usage of eBPF
 - Rich experience in distributed systems and distributed machine learning frameworks
 - Strong engineering development experience, mastering various compiler-related tools and static analysis tools
- **Physics & Math:**
 - Highly skilled in Computational Fluid Dynamics and Molecular Dynamics
 - Strong knowledge in numerical methods and linear algebra
 - Proficient in Quantum Mechanics and Quantum Electrodynamics, with some knowledge of quantum computing

SPECIAL EXPERIENCE

- **Open Source Enthusiast:** Contributing to notable projects like GNOME, LAMMPS, and LLVM, I've used GitHub since 2019 to showcase my development journey and ideas.
- **Running a Science-Themed Cafe:** Leveraging software development income, I established Quantum Coffee near my school—a collaborative space encouraging students to explore and discuss scientific topics across disciplines.
- **UNICEF Charity Projects:** I have donated 10% of all personal income to children's charities since 2019.